Linear Algebra II, Sheet 1, HT26

Determinants.

Main course

1. Verify directly that $\det AB = \det A \times \det B = \det BA$ where

$$A = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \\ 2 & 0 & 1 \end{pmatrix} \qquad \text{and} \qquad B = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & -1 \\ 2 & 0 & 3 \end{pmatrix}.$$

2. Use EROs and ECOs to evaluate the following 4×4 determinants.

$$\begin{vmatrix} 1 & 2 & 3 & 1 \\ -2 & 0 & 0 & 2 \\ 6 & 1 & 3 & 4 \\ 0 & 1 & 3 & -1 \end{vmatrix}, \qquad \begin{vmatrix} 1 & 2 & 4 & 2 \\ 2 & -3 & 1 & -3 \\ 7 & -1 & 0 & 1 \\ 1 & -2 & 1 & 0 \end{vmatrix}.$$

3. Show, for any real number x, that

$$\begin{vmatrix} x^2 & (x+1)^2 & (x+2)^2 \\ (x+1)^2 & (x+2)^2 & (x+3)^2 \\ (x+2)^2 & (x+3)^2 & (x+4)^2 \end{vmatrix} = -8 \qquad \text{and} \qquad \begin{vmatrix} 1 & 1 & 1 & 1 \\ x & 1 & 1 & 1 \\ x & x & 1 & 1 \\ x & x & x & 1 \end{vmatrix} = (1-x)^3.$$

4. Let A be the $(m+n) \times (m+n)$ matrix below where U, V, W are respectively $m \times m$, $m \times n$, $n \times n$ matrices.

$$A = \begin{pmatrix} U & V \\ 0 & W \end{pmatrix}.$$

Show that

$$\det A = \det U \times \det W.$$

5. (*Vandermonde matrix*) For $n \geqslant 2$ and real numbers $x_1, \ldots, x_n$ we define

$$V_n = \begin{pmatrix} 1 & x_1 & x_1^2 & \cdots & x_1^{n-1} \\ 1 & x_2 & x_2^2 & \cdots & x_2^{n-1} \\ 1 & x_3 & x_3^2 & \cdots & x_3^{n-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_n & x_n^2 & \cdots & x_n^{n-1} \end{pmatrix}.$$

(i) Show that

$$\det V_n = \prod_{i>j} (x_i - x_j).$$

[Hint: begin by adding $-x_1 \times$ (kth column) to $(k+1)$th column for $1 \leqslant k < n$.]

(ii) Deduce that V_n is invertible if and only if the x_i are distinct.

(iii) Let $x_0, x_1, \ldots, x_n, y_0, y_1, \ldots, y_n$ be real numbers with the x_i distinct. Show that there is a unique polynomial $p(x)$ of degree n or less such that $p(x_i) = y_i$ for each $0 \leqslant i \leqslant n$. Finding such $p(x)$ is called *Lagrangian interpolation*.

(iv) Determine $p(x)$ when $y_0 = 1$ and $y_i = 0$ for $i \geqslant 1$.

Starter

S1. Give a counter-example to $\det(A+B) = \det A + \det B$.

S2. Show that an orthogonal matrix has determinant 1 or -1. Is the converse true?

S3. Prove, by induction, that the determinant of a triangular matrix is the product of its diagonal entries.

Pudding

P1. In the case of the determinant

$$\begin{vmatrix} 1 & -3 & 2 \\ 0 & 7 & 1 \\ -5 & 1 & 3 \end{vmatrix}$$

verify that the following three expressions

$$a_{21}C_{21} + a_{22}C_{22} + a_{23}C_{23}, \qquad a_{12}C_{12} + a_{22}C_{22} + a_{32}C_{32}, \qquad a_{13}C_{13} + a_{23}C_{23} + a_{33}C_{33},$$

all agree with the given determinant. Here C_{ij} denotes the (i,j)th cofactor.

P2. Let A be a square matrix. Show that if A has a real square root then $\det A \geqslant 0$. Is the converse true?

P3. Define the $n \times n$ determinant D_n by

$$D_n = \begin{vmatrix} 1 & -1 & 0 & 0 & \cdots & 0 \\ 1 & 1 & -1 & 0 & \cdots & 0 \\ 0 & 1 & 1 & -1 & \ddots & \vdots \\ 0 & 0 & 1 & \ddots & \ddots & 0 \\ \vdots & \vdots & \ddots & \ddots & 1 & -1 \\ 0 & 0 & \cdots & 0 & 1 & 1 \end{vmatrix}.$$

Show that $D_1 = 1$ and $D_2 = 2$ and that $D_{n+2} = D_{n+1} + D_n$ for $n \geqslant 1$. Deduce that D_n is the $(n+1)$th Fibonacci number.